INTRODUCTION TO ANALYSIS OF ALGORITHMS

Data Structures and Algorithm

Outline

- Introduction to Algorithms
- Example Problem and Algorithms
- Analysis of Algorithms
- Runtime Analysis
- Exercise

Introduction

- A famous quote: Program = Algorithm + Data Structure.
- All of you have programmed; thus have already been exposed to algorithms and data structure.
- Perhaps you didn't see them as separate entities;
- Perhaps you saw data structures as simple programming constructs (provided by STL--standard template library).
- However, data structures are quite distinct from algorithms, and very important in their own right.

Example Algorithms

- Two algorithms for computing the Factorial
- Which one is better?

- ```
  int factorial (int n) {
    if (n <= 1)    return 1;
    else   return n * factorial(n-1);
  }
  ```

- ```
  int factorial (int n) {
    if (n<=1)     return 1;
    else {
      fact = 1;
      for (k=2; k<=n; k++)
          fact *= k;
      return fact;
    }
  }
  ```

Examples of famous algorithms

- Constructions of Euclid
- Newton's root finding
- Fast Fourier Transform
- Compression (Huffman, Lempel-Ziv, GIF, MPEG)
- DES, RSA encryption
- Simplex algorithm for linear programming
- Shortest Path Algorithms (Dijkstra, Bellman-Ford)
- Error correcting codes (CDs, DVDs)
- TCP congestion control, IP routing
- Pattern matching (Genomics)
- Search Engines

Role of Algorithms in Modern World

- Enormous amount of data
  - E-commerce (Amazon, Ebay)
  - Network traffic (telecom billing, monitoring)
  - Database transactions (Sales, inventory)
  - Scientific measurements (astrophysics, geology)
  - Sensor networks. RFID tags
  - Bioinformatics (genome, protein bank)

- Amazon hired first *Chief Algorithms Officer* (Udi Manber)

Max Subsequence Problem

- Given a sequence of integers A1, A2, …, An, find the maximum possible value of a subsequence Ai, …, Aj.
- Numbers can be negative.
- You want a contiguous chunk with largest sum.
- Example: -2, 11, -4, 13, -5, -2
- The answer is 20 (subseq. A2 through A4).

- We will discuss 4 different algorithms, with time complexities $O(n^3)$, $O(n^2)$, $O(n \log n)$, and $O(n)$.
- With $n = 10^6$, algorithm 1 may take > 10 years; algorithm 4 will take a fraction of a second!

Algorithm 1 for Max Subsequence Sum

- Given $A_1,\ldots,A_n$, find the maximum value of $A_i+A_{i+1}+\cdots+A_j$
  0 if the max value is negative

```
int maxSum = 0;

for( int i = 0; i < a.size( ); i++ )
for( int j = i; j < a.size( ); j++ )
{
        int thisSum = 0;
        for( int k = i; k <= j; k++ )
                thisSum += a[ k ];
        if( thisSum > maxSum )
                maxSum   = thisSum;
}
return maxSum;
```

$O(1)$

$O(1)$

$O(1)$

$O(1)$

$O(j-i)$

$O(\sum_{j=i}^{n-1}(j-i))$

$O(\sum_{i=0}^{n-1}\sum_{j=i}^{n-1}(j-i))$

- Time complexity: $O(n^3)$

Algorithm 2

- Idea: Given sum from i to j-1, we can compute the sum from i to j in constant time.
- This eliminates one nested loop, and reduces the running time to $O(n^2)$.

```
into maxSum = 0;

for( int i = 0; i < a.size( ); i++ )
        int thisSum = 0;
        for( int j = i; j < a.size( ); j++ )
        {
            thisSum += a[ j ];
            if( thisSum > maxSum )
                maxSum   = thisSum;
        }
return maxSum;
```

Algorithm 3

- This algorithm uses divide-and-conquer paradigm.
- Suppose we split the input sequence at midpoint.
- The max subsequence is entirely in the left half, entirely in the right half, or it straddles the midpoint.
- Example:

left half	right half
4 -3 5 -2	-1 2 6 -2

- Max in left is 6 (A1 through A3); max in right is 8 (A6 through A7). But straddling max is 11 (A1 thru A7).

Algorithm 3 (cont.)

- Example:

left half					right half			
4	-3	5	-2	|	-1	2	6	-2

- Max subsequences in each half found by recursion.
- How do we find the straddling max subsequence?
- Key Observation:
  - Left half of the straddling sequence is the max subsequence ending with -2.
  - Right half is the max subsequence beginning with -1.

- A linear scan lets us compute these in O(n) time.

Algorithm 3: Analysis

- The divide and conquer is best analyzed through recurrence:

$$T(1) = 1$$
$$T(n) = 2T(n/2) + O(n)$$

- This recurrence solves to $T(n) = O(n \log n)$.

Algorithm 4

2, 3, -2, 1, -5, 4, 1, -3, 4, -1, 2

```
    int maxSum = 0, thisSum = 0;

    for( int j = 0; j < a.size( ); j++ )
    {
        thisSum += a[ j ];

        if ( thisSum > maxSum )
            maxSum = thisSum;
        else if ( thisSum < 0 )
            thisSum = 0;
    }
    return maxSum;
}
```

- Time complexity clearly $O(n)$
- But why does it work? I.e. proof of correctness.

Proof of Correctness

- Max subsequence cannot start or end at a negative Ai.
- More generally, the max subsequence cannot have a prefix with a negative sum.

  Ex: -2 11 -4 13 -5 -2
- Thus, if we ever find that Ai through Aj sums to < 0, then we can advance i to j+1
  - Proof. Suppose j is the first index after i when the sum becomes < 0
  - The max subsequence cannot start at any p between i and j. Because A_i through A_{p-1} is positive, so starting at i would have been even better.

Algorithm 4

```
int maxSum = 0, thisSum = 0;

for( int j = 0; j < a.size( ); j++ )
{
        thisSum += a[ j ];

        if ( thisSum > maxSum )
                maxSum = thisSum;
        else if ( thisSum < 0 )
                thisSum = 0;
}
return maxSum
```

- The algorithm resets whenever prefix is < 0. Otherwise, it forms new sums and updates maxSum in one pass.

Why Efficient Algorithms Matter

- Suppose $N = 10^6$
- A PC can read/process N records in 1 sec.
- But if some algorithm does N*N computation, then it takes 1M seconds = 11 days!!!

- 100 City Traveling Salesman Problem.
  - A supercomputer checking 100 billion tours/sec still requires 10^{100} years!

- Fast factoring algorithms can break encryption schemes. Algorithms research determines what is safe code length. (> 100 digits)

How to Measure Algorithm Performance

- What metric should be used to judge algorithms?
  - Length of the program (lines of code)
  - Ease of programming (bugs, maintenance)
  - Memory required
  - ❑Running time

- **Running time is the dominant standard.**
  - Quantifiable and easy to compare
  - Often the critical bottleneck

Abstraction

- An algorithm may run differently depending on:
  - the hardware platform (PC, Cray, Sun)
  - the programming language (C, Java, C++)
  - the programmer (you, me, Bill Gates)

- While different in detail, all hardware and prog models are equivalent in some sense: **Turing machines**.

- It suffices to count basic operations.

- Crude but valuable measure of algorithm's performance *as a function of input size.*

Average, Best, and Worst-Case

- On which input instances should the algorithm's performance be judged?

- Average case:
  - Real world distributions difficult to predict
- Best case:
  - Seems unrealistic
- Worst case:
  - Gives an absolute guarantee
  - **We will use the worst-case measure.**

Examples

- Vector addition **Z = A+B**

```
for (int i=0; i<n; i++)
    Z[i] = A[i] + B[i];
```

$T(n) = c\ n$

- Vector (inner) multiplication **z =A*B**

```
z = 0;
for (int i=0; i<n; i++)
    z = z + A[i]*B[i];
```

$T(n) = c' + c_1\ n$

Examples

- Vector (outer) multiplication $Z = A*B^T$

```
for (int i=0; i<n; i++)
    for (int j=0; j<n; j++)
        Z[i,j] = A[i] * B[j];
```

$T(n) = c_2 n^2;$

- A program does all the above

$T(n) = c_0 + c_1 n + c_2 n^2;$

Simplifying the Bound

- $T(n) = c_k\, n^k + c_{k-1}\, n^{k-1} + c_{k-2}\, n^{k-2} + \ldots + c_1\, n + c_o$
  - too complicated
  - too many terms
  - Difficult to compare two expressions, each with 10 or 20 terms
- Do we really need that many terms?

Simplifications

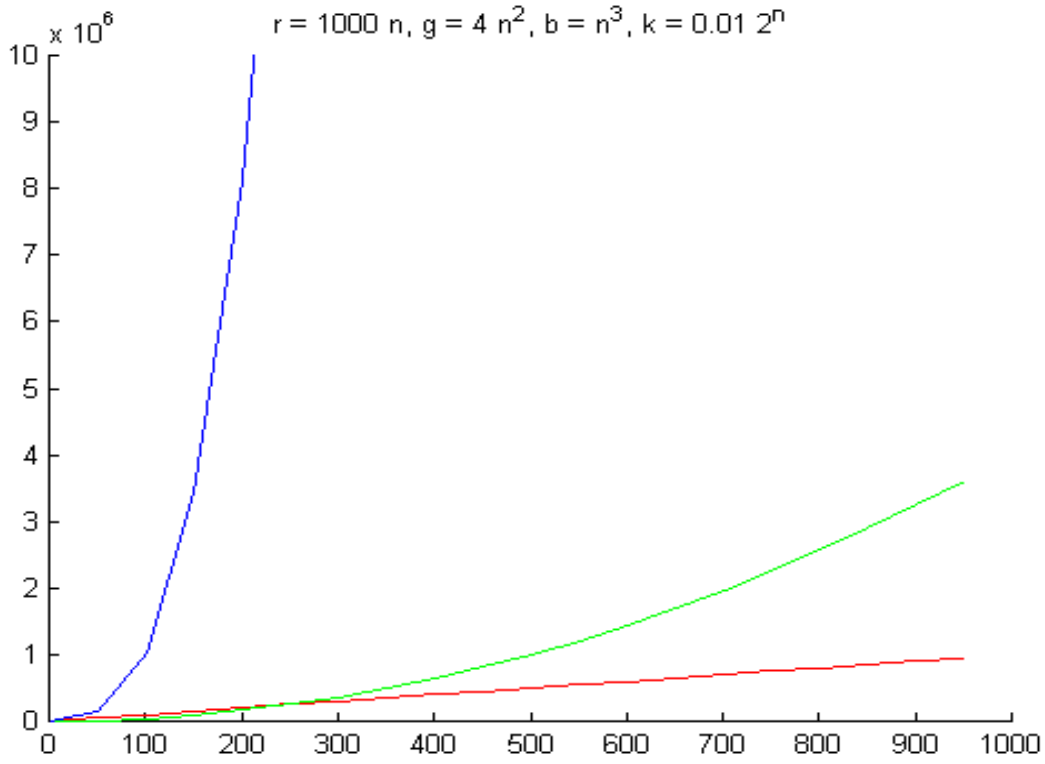
- Keep just one term!
  - the fastest growing term (dominates the runtime)
- No constant coefficients are kept
  - Constant coefficients affected by machines, languages, etc.

- **Asymtotic behavior** (as *n* gets large) is determined entirely by the **leading** term.

  - Example. $T(n) = 10\,n^3 + n^2 + 40n + 800$
    - *If n = 1,000, then T(n) = 10,001,040,800*
    - error is 0.01% if we drop all but the n^3 term
  - In an assembly line the slowest worker determines the throughput rate

Simplification

- Drop the constant coefficient
  - Does not effect the relative order



Simplification

- The *faster* growing term (such as 2^n) *eventually* will outgrow the slower growing terms (e.g., 1000 n) no matter what their coefficients!

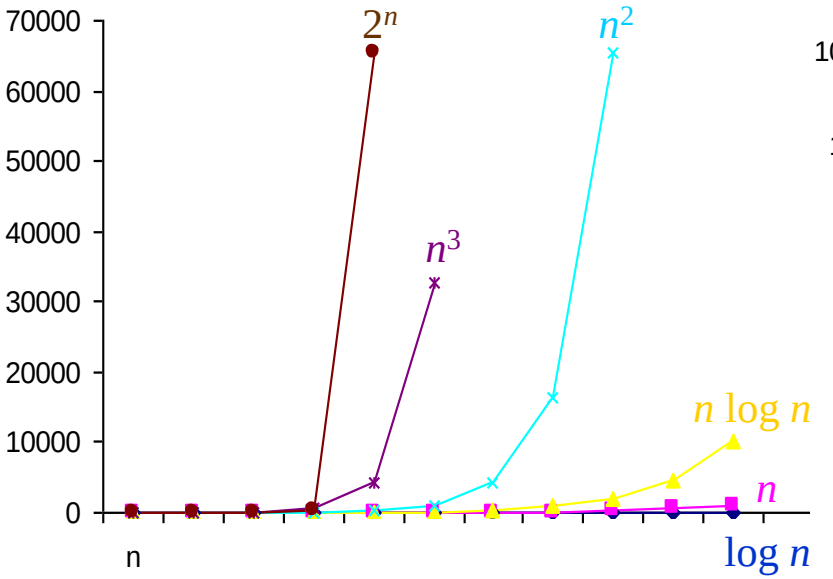
- Put another way, given a certain increase in allocated time, a higher order algorithm will not reap the benefit by solving much larger problem

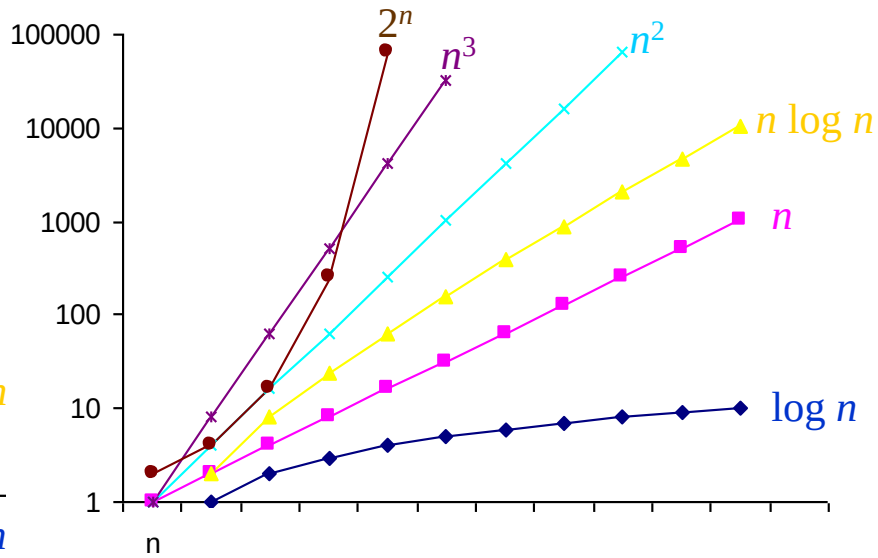
Complexity and Tractability

	$T(n)$						
n	n	$n \log n$	n^2	n^3	n^4	n^{10}	2^n
10	.01μs	.03μs	.1μs	1μs	10μs	10s	1μs
20	.02μs	.09μs	.4μs	8μs	160μs	2.84h	1ms
30	.03μs	.15μs	.9μs	27μs	810μs	6.83d	1s
40	.04μs	.21μs	1.6μs	64μs	2.56ms	121d	18m
50	.05μs	.28μs	2.5μs	125μs	6.25ms	3.1y	13d
100	.1μs	.66μs	10μs	1ms	100ms	3171y	4×10^{13}y
10^3	1μs	9.96μs	1ms	1s	16.67m	3.17×10^{13}y	32×10^{283}y
10^4	10μs	130μs	100ms	16.67m	115.7d	3.17×10^{23}y	
10^5	100μs	1.66ms	10s	11.57d	3171y	3.17×10^{33}y	
10^6	1ms	19.92ms	16.67m	31.71y	3.17×10^{7}y	3.17×10^{43}y	

Assume the computer does 1 billion ops per sec.

$\log n$	n	$n \log n$	n^2	n^3	2^n
0	1	0	1	1	2
1	2	2	4	8	4
2	4	8	16	64	16
3	8	24	64	512	256
4	16	64	256	4096	65,536
5	32	160	1,024	32,768	4,294,967,296





Another View

- More resources (time and/or processing power) translate into large problems solved if complexity is low

T(n)	Problem size solved in 10^3 sec	Problem size solved in 10^4 sec	Increase in Problem size
100n	10	100	10
1000n	1	10	10
$5n^2$	14	45	3.2
N^3	10	22	2.2
2^n	10	13	1.3

Asympotics

T(n)	keep one	drop coef
$3n^2+4n+1$	$3\ n^2$	n^2
$101\ n^2+102$	$101\ n^2$	n^2
$15\ n^2+6n$	$15\ n^2$	n^2
$a\ n^2+bn+c$	$a\ n^2$	n^2

- They all have the same "growth" rate

Caveats

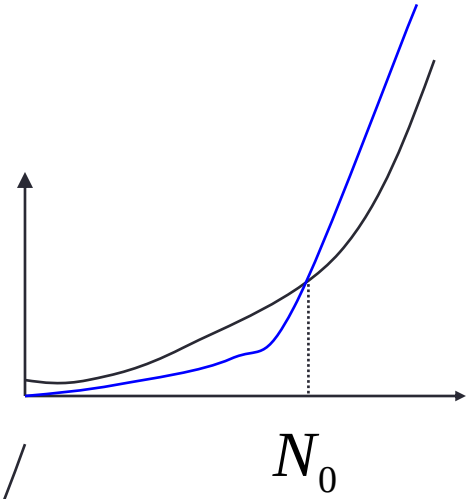
- Follow the spirit, not the letter
  - a 100n algorithm is more expensive than n^2 algorithm when $n < 100$
- Other considerations:
  - a program used only a few times
  - a program run on small data sets
  - ease of coding, porting, maintenance
  - memory requirements

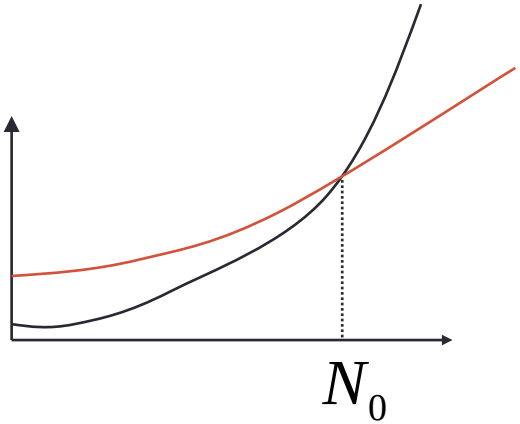
Asymptotic Notations

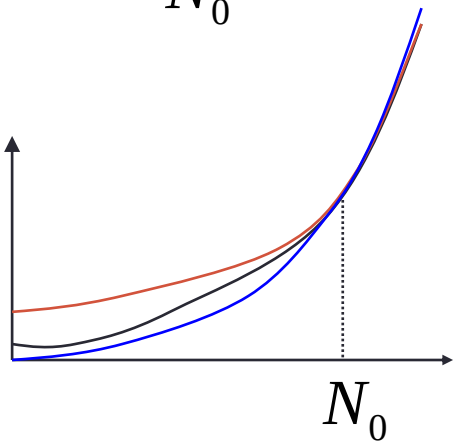
- Big-O, "bounded above by": $T(n) = O(f(n))$
  - For some c and N, $T(n) \leq c \cdot f(n)$ whenever $n > N$.
- Big-Omega, "bounded below by": $T(n) = \Omega(f(n))$
  - For some $c>0$ and N, $T(n) \geq c \cdot f(n)$ whenever $n > N$.
  - Same as $f(n) = O(T(n))$.
- Big-Theta, "bounded above and below": $T(n) = \Theta(f(n))$
  - $T(n) = O(f(n))$ and also $T(n) = \Omega(f(n))$
- Little-o, "strictly bounded above": $T(n) = o(f(n))$
  - $T(n)/f(n) \rightarrow 0$ as $n \rightarrow \infty$

By Pictures

- Big-Oh (most commonly used)
  - bounded above
- Big-Omega
  - bounded below
- Big-Theta
  - exactly
- Small-o
  - not as expensive as ...







Example

$$T(n) = n^3 + 2n^2$$

$O(?)$	$\Omega(?)$
∞	0
n^{10}	n
n^5	n^2
n^3	n^3

Examples

$f(n)$	Asymptomic
c	$\Theta(1)$
$\sum_{i=1}^{k} c_i n^i$	$\Theta(n^k)$
$\sum_{i=1}^{n} i$	$\Theta(n^2)$
$\sum_{i=1}^{n} i^2$	$\Theta(n^3)$
$\sum_{i=1}^{n} i^k$	$\Theta(n^{k+1})$
$\sum_{i=0}^{n} r^i$	$\Theta(r^n)$
$n!$	$\Theta(n(n/e)^n)$
$\sum_{i=1}^{n} 1/i$	$\Theta(\log n)$

Summary (Why O(n)?)

- $T(n) = c_k n^k + c_{k-1} n^{k-1} + c_{k-2} n^{k-2} + \ldots + c_1 n + c_o$
- Too complicated
- $O(n^k)$
  - a single term with constant coefficient dropped
- Much simpler, extra terms and coefficients *do not matter* asymptotically
- Other criteria hard to quantify

Runtime Analysis

- Useful rules
  - simple statements (read, write, assign)
    - $O(1)$ (constant)
  - simple operations (+ - * / == > >= < <=)
    - $O(1)$
  - sequence of simple statements/operations
    - rule of sums
  - for, do, while loops
    - rules of products

Runtime Analysis (cont.)

- Two important rules
  - Rule of sums
    - if you do a number of operations in sequence, the runtime is dominated by the most expensive operation
  - Rule of products
    - if you repeat an operation a number of times, the total runtime is the runtime of the operation multiplied by the iteration count

Runtime Analysis (cont.)

if (cond) then	$O(1)$
$\quad$ $body_1$	$T_1(n)$
else	
$\quad$ $body_2$	$T_2(n)$
endif	

$$T(n) = O(\max(T_1(n), T_2(n)))$$

Runtime Analysis (cont.)

- Method calls
  - A calls B
  - B calls C
  - etc.
- A sequence of operations when call sequences are flattened

$T(n) = max(T_A(n), T_B(n), T_C(n))$

Example

```
for (i=1; i<n; i++)
    if A(i) > maxVal then
        maxVal= A(i);
        maxPos= i;
```

Asymptotic Complexity: O(n)

Example

```
for (i=1; i<n-1; i++)
    for (j=n; j>= i+1;  j--)
        if (A(j-1) > A(j)) then
            temp = A(j-1);
            A(j-1) = A(j);
            A(j) = tmp;
        endif
    endfor
endfor
```

- Asymptotic Complexity is $O(n^2)$

Run Time for Recursive Programs

- T(n) is defined recursively in terms of T(k), k<n
- The recurrence relations allow T(n) to be "unwound" recursively into some base cases (e.g., T(0) or T(1)).
- Examples:
  - Factorial
  - Hanoi towers

Example: Factorial

```
int factorial (int n) {
   if (n<=1)   return 1;
   else  return n * factorial(n-1);
}
```

factorial (n) = n*n-1*n-2* … *1

n * factorial(n-1)

n-1 * factorial(n-2)

n-2 * factorial(n-3)

…

2 *factorial(1)

$$
\begin{aligned}
&T(n) \\
&= T(n-1) + d \\
&= T(n-2) + d + d \\
&= T(n-3) + d + d + d \\
&= \ldots. \\
&= T(1) + (n-1) * d \\
&= c + (n-1) * d \\
&= O(n)
\end{aligned}
$$

T(n)

T(n-1)

T(n-2)

T(1)

Example: Factorial (cont.)

```
int factorial1(int n) {
   if (n<=1)  return 1;
   else {
     fact = 1;
     for (k=2;k<=n;k++)
         fact *= k;
     return fact;
   }
}
```

$O(1)$

$O(1)$

$O(n)$

- Both algorithms are O(n).

Example: Hanoi Towers

- Hanoi(n,A,B,C) =
  Hanoi(n-1,A,C,B)+Hanoi(1,A,B,C)+Hanoi(n-1,C,B,A)

$$
\begin{aligned}
&T(n) \\
&= 2T(n-1)+c \\
&= 2^2T(n-2)+2c+c \\
&= 2^3T(n-3)+2^2c+2c+c \\
&= .... \\
&= 2^{n-1}T(1)+(2^{n-2}+...+2+1)c \\
&= (2^{n-1}+2^{n-2}+...+2+1)c \\
&= O(2^n)
\end{aligned}
$$

Exercise

Write program and analyze complexity:

- Perform polynomial additional and multiplication
- Determine the 2nd largest element in an array – the kth largest element in an array.
- Find longest increasing subsequence:
  - contiguous case: 2 algorithms with different complexity

    0, 8, <u>4, 12, 24,</u> 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15
  - uncontiguous case:

    <u>0</u>, 8, 4, 12, 2, <u>4</u>, 10, <u>6</u>, 14, 1, <u>9</u>, 5, <u>13</u>, 3, 11, 7, <u>15</u>